O-RAN.WG3.O1-Interface-for-NearRT-RIC-R003-v01.00

Technical Specification

O-RAN Work Group 3
O1 Interface Specification for Near Real Time RAN Intelligent Controller

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Contents

Foreword

This Technical Specification (TS) has been produced by the O-RAN Alliance.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

yy: the second digit-group is incremented when editorial only changes have been incorporated in the document. Always 2 digits with leading zero if needed.

zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

The present document defines O-RAN OAM interface functions and protocols for the O-RAN O1 interface for the Near RT RIC. The document studies the functions conveyed over the interface, including management functions, procedures, operations, and corresponding solutions, and identifies existing standards and industry work that can serve as a basis for O-RAN work.

This document will follow the requirements specification language defined in IETF RFC 2119 [33] updated by RFC 8174 [37]. For consistency requirements are specified using “SHALL” to indicate that the implementation is required.

2 References

2.1 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[1] 3GPP TR 21.905: “Vocabulary for 3GPP Specifications”

[2] O-RAN-WG3.E2GAP, “O-RAN Working Group 3, Near-Real-time RAN Intelligent Controller, E2 General Aspects and Principles”.

[3] O-RAN-WG3.E2AP, “O-RAN Working Group 3, Near-Real-time RAN Intelligent Controller, E2 Application Protocol (E2AP)”.

[4] O-RAN-WG1.OAM Architecture, “O-RAN Operations and Maintenance Architecture”.

[5] O-RAN-WG1.O1-Interface, “O-RAN Operations and Maintenance Interface Specification”.

[6] 3GPP TS 33.401: “3GPP System Architecture Evolution (SAE); Security architecture”.

[7] 3GPP TS 33.501: “Security architecture and procedures for 5G System”.

[8] O-RAN-WG2.A1.GA&P, “O-RAN Working Group 2, A1 interface: General Aspects and Principles”.

[9] O-RAN-WG2.A1AP, "O-RAN Working Group 2, A1 Interface: Application Protocol".

[10] O-RAN-WG1.O-RAN Architecture, "O-RAN Working Group 1, O-RAN Architecture Description".

[11] 3GPP TS 36.401: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); Architecture Description".

[12] 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description; Stage 2".

[13] 3GPP TS 37.324: "Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Service Data Adaptation Protocol (SDAP) specification".

[14] 3GPP TS 38.331: "NR; Radio Resource Control (RRC); Protocol specification".

[15] 3GPP TS 38.323: "NR; Packet Data Convergence Protocol (PDCP) specification".

[16] 3GPP TS 38.322: "NR; Radio Link Control (RLC) protocol specification".

[17] 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification".

[18] 3GPP TS 38.201: "NR; Physical layer; General description".

[19] O-RAN WG10: O-RAN Information Model and Data Models Specification,

[20] O-RAN WG3.RICARCH-v02.00

[21] 3GPP TS 28.622: "Generic Network Resource Mode (NRM) "

[22] O-RAN WG1.Information Model and Data Models-v01.00

[23] AlarmDictionaryModel-2021-07.11-13645

[24] 3GPP TS 28.541: "Management and orchestration; 5G Network Resource Model (NRM)"

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms [given in [i.1] and the following] apply:

N/A

3.2 Symbols

For the purposes of the present document, the symbols [given in [i.1] and the following] apply:

N/A

3.3 Abbreviations

For the purposes of the present document, the abbreviations [given in [i.1] and the following] apply:

FCAPS	Fault, Configuration, Accounting, Performance, Security
FM	Fault Management
IOC	Information Object Class
ME	Managed Element

MF	Managed Function
MnS	Management Service
MO	Managed Object
MOI	Managed Object Instance
Near-RT RIC	O-RAN Near Real Time RAN Intelligent Controller
NETCONF	NETwork CONFiguration protocol
NF	Network Function
Non-RT RIC	O-RAN Non-Real Time RAN Intelligent Controller
PM	Performance Management or Performance Measurements
SMO	Service Management and Orchestration
TR	Technical Report

4 Management Services

4.1 Provisioning Management Services

Provisioning Management Services MnS for Near RT RIC platform shall be aligned with [5], Chapter 2.1

4.2 Fault Supervision Management Services

Fault Supervision MnS for Near RT RIC platform shall be aligned with [5], Chapter 2.2

4.2.1 Requirements

Fault Management for the Near-RT RIC Platform is based on the O-RAN-WG1.O1-Interface, "O-RAN Operations and Maintenance Interface Specification [5]
The relevant sections are:

- 4.1.10 Subscription Control
- 4.2.1 Fault Notification
- 4.2.2 Fault Supervision Control

4.3 Performance Assurance Management Services

Performance Assurance Management Services MnS for Near RT RIC platform shall be aligned with [5], Chapter 2.3

4.4 Trace Management Services

5 Model

5.1 Imported and associated information entities

5.1.1 Imported information entities and local labels

Label reference	Local label
TS 28.622 [21], IOC, ManagedFunction	ManagedFunction
TS 28.622 [21], IOC, EP_RP	EP_RP
O-RAN.WG10.Information Model and Data Models	ManagedApplication

5.1.2 Associated information entities and local labels

Label reference	Local label
TS 28.622 [21], IOC, ManagedElement	ManagedElement

5.2 Class Diagrams

5.2.1 Relationships

```
@startuml
skinparam ClassStereotypeFontStyle normal
hide empty members

class        ManagedElement  <<InformationObjectClass>>
class       "NearRTRICFunction" as ric <<InformationObjectClass>>
class       "EP_E2" as E2 <<InformationObjectClass>>
class        "E2 Node" as bts   <<proxyClass>>

ric "*" -d-* "1" ManagedElement : <<names>>
ric "1" *-right-"*" E2 : <<names>>

E2 "*" -right-> "    0..1" bts : "       "
note top of bts
Proxy class representing
any of the following E2 EndPoints:
O-CU-CP
O-CU-UP
O-DU
O-eNB
end note
@enduml
```

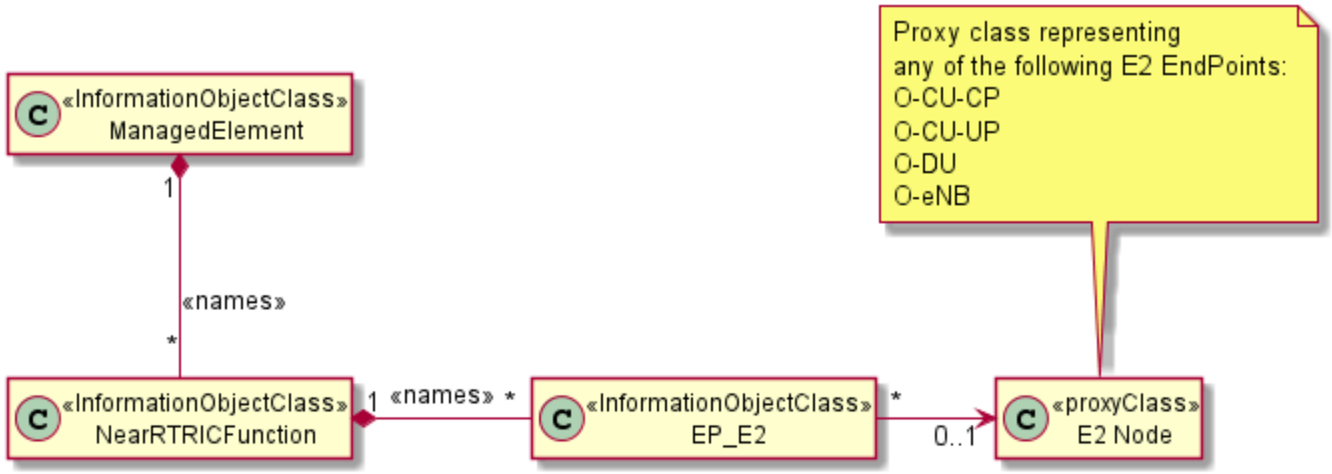


Figure 5.2.1-1: Near RT RIC Containment

Further disaggregation of the internal NearRTRICFunction management is FFS.

5.2.2 Inheritance

```
@startuml
hide empty members
skinparam ClassStereotypeFontStyle normal
class NearRTRICFunction <<InformationObjectClass>>
Abstract ManagedFunction <<InformationObjectClass>>
ManagedFunction  <|--  NearRTRICFunction

@enduml
```

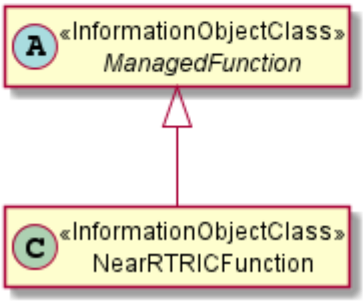


Figure 5.2.2-1: Near-RT RIC Function inheritance

```
@startuml
hide empty members
skinparam ClassStereotypeFontStyle normal
abstract class EP_RP <<InformationObjectClass>>
class EP_E2 <<InformationObjectClass>>
EP_RP  <|--  EP_E2

@enduml
```

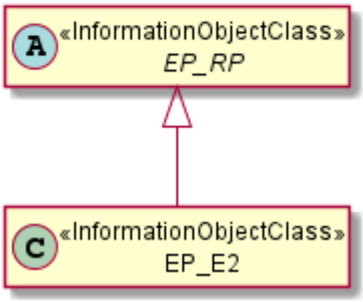


Figure 5.2.2-2: EP_E2 Inheritance

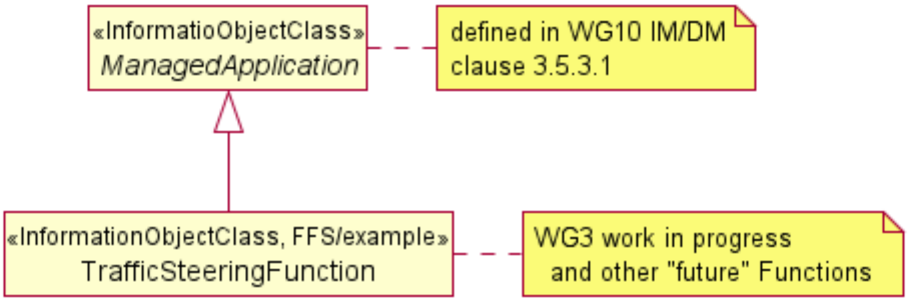
```
@startuml xApp inheritance (2022-11-04)

skin rose
skinparam ClassStereotypeFontStyle normal
skinparam style strictuml
hide empty members

abstract class ManagedApplication <<InformatioObjectClass>>
class TrafficSteeringFunction <<InformationObjectClass, FFS/example>>
ManagedApplication <|-d- TrafficSteeringFunction

note right of ManagedApplication
  defined in WG10 IM/DM
  clause 3.5.3.1
  end note

note right of TrafficSteeringFunction
  WG3 work in progress
    and other "future" Functions
  end note
@enduml
```



NOTE: This figure will be removed in the next version of the current document and replaced by one or more concrete xApp Use Cases

Figure 5.2.2-2: Example of inheritance from *ManagedApplication*

5.3 Class Definitions

5.3.1 NearRTRICFunction

5.3.1.1 Definition

The NearRTRICFunction IOC represents the Management aspects of the aggregated functions making up the Near-Rt RIC defined in clause 4.3.2 of O-RAN.WG1.O-RAN-Architecture-Description.

5.3.1.2 Attributes

The NearRTRICFunction IOC includes the attributes below and those attributes inherited through *ManagedFunction* IOC (defined in TS 28.622 [21]).

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
PLMN Identity (3GPP 38.423 Clause 9.2.2.4)	M	T	T	F	T
Near-RT RIC ID	M	T	T	F	T
Attribute related to role					
applicationDNList	M	T	T	F	T

5.3.1.3 Attribute constraints

None

5.3.1.4 Notifications

None

5.3.2 EP_E2

The common notifications defined in clause 4.5 in TS 28.622 [21] are valid for this IOC

5.3.2.1 Definition

The EP-E2 IOC represents the Management aspects of the E2T defined in RICArch [4] section 6.2.

5.3.2.2 Attributes

The EP_E2 IOC includes the attributes below and those attributes inherited through EP_RP IOC (defined in TS 28.622 [21])

Attribute Name	S	isReadable	isWritable	isInvariant	isNotifyable
tRICEventCreate	O	T	T	F	T
tRICEventDelete	O	T	T	F	T
tRICControl	O	T	T	F	T

5.3.2.3 Attribute constraints

None

5.3.2.4 Notifications

None

5.5 Attribute definitions

5.5.1 Attribute properties

Attribute Name	Documentation and Allowed Values	Properties
tRICEventCreate	Defined in O-RAN.WG3.E2AP Clause 9.5 Specifies the maximum time for the RIC Subscription Request event creation procedure in the E2 Node. allowedValues: [0..4095] ms	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: True
tRICEventelete	Defined in O-RAN.WG3.E2AP Clause 9.5 Specifies the maximum time for the RIC Subscription Request event deletion procedure in the E2 Node. allowedValues: [0..4095] ms	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: True
tRICControl	Defined in O-RAN.WG3.E2AP Clause 9.5 Specifies the maximum time for the RIC Control Request event request procedure in the E2 Node. allowedValues: [0..4095] ms	type: Integer multiplicity: 1 isOrdered: N/A isUnique: N/A defaultValue: Null isNullable: True
applicationDNList	This attribute contains the DNs of the hosted applications (MOIs of the concrete IOCs inheriting from ManagedApplication). allowedValues: Not applicable.	type: DN multiplicity: 0..* isOrdered: False isUnique: True defaultValue: None isNullable: False
nearRTRICID	Defined in O-RAN.WG3.E2AP-v02.01 Clause 9.2.4	
pLMNIdentity	Defined in 3GPP 38.423 Clause 9.2.2.4	

Revision history

Date	Revision	Description
2021.09.27	00.00.01	First draft of O-RAN OAM Interface Specification for Near RT RIC
2021.11.05	00.00.02	Incorporating CRs NOK0001 and NOK0002
2022.01.08	00.00.03	Document updated to incorporate CRs NOK0003 and CR NOK0007 rev8
2022.07.20	00.00.04	New Baseline
2022.11.09	00.00.05	Baseline for November train
2022.11.17	00.00.06	Addressing review comments
2022.11.18	01.00	Candidate for November train

